

Research on Advanced 2D Virtual Try-On and Dynamic Garment Alignment

AI/ML Internship — Day 10

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Introduction

Virtual Try-On systems aim to provide users with a realistic digital fitting experience. Traditional 2D try-on methods simply overlay a clothing image onto a user's body, often producing unrealistic results due to incorrect scaling and positioning.

Modern computer vision techniques improve this process by introducing dynamic garment scaling, pose-aware alignment, and transparent garment integration. These improvements help garments adapt more naturally to different body shapes and user poses.

This research focuses on studying advanced 2D Virtual Try-On methods and the technologies used to improve garment fitting accuracy.

Objectives

- Study advanced 2D Virtual Try-On systems.
- Understand dynamic garment scaling.
- Analyse pose-based garment alignment.
- Explore transparent PNG garments.
- Study cloth segmentation support.

1. Traditional 2D Virtual Try-On

Basic systems perform:

- User image loading
- Cloth image loading
- Fixed resizing
- Simple overlay

Limitations:

- Poor alignment
- Incorrect scaling
- Unrealistic appearance

2. Dynamic Garment Scaling

Dynamic scaling automatically adjusts garment size according to body proportions.

Factors considered:

- Shoulder width
- Torso height
- Body width

Benefits:

- Better fitting
 - Improved realism
 - Reduced placement errors
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3. Pose-Based Alignment

Body landmarks are detected before placing garments.

Important landmarks:

- Left Shoulder
- Right Shoulder
- Chest Centre
- Waist Centre

Using pose estimation significantly improves garment positioning.

4. Transparent PNG Garments

Transparent PNG clothing assets remove unwanted backgrounds.

Advantages:

- Cleaner overlay
 - Better visual quality
 - Easier image blending
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5. Cloth Segmentation Support

Segmentation helps AI identify clothing regions automatically.

Benefits:

- Accurate placement
 - Better body understanding
 - Reduced overlap errors
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6. Multi-Image Testing

Testing should include:

- Different body sizes
- Multiple poses
- Various lighting conditions

This improves system reliability.

7. Comparison Between Basic and Advanced Systems

Feature	Basic Try-On	Advanced Try-On
Scaling	Fixed	Dynamic
Alignment	Manual	Pose-Based
Clothing	Normal Image	Transparent PNG
Blending	Direct Overlay	Alpha Blending
Testing	Limited	Multi-image

8. Research Findings

Dynamic garment scaling and pose-based alignment significantly improve Virtual Try-On quality. Transparent PNG garments and cloth segmentation produce more natural outputs.

9. Learning Outcomes

- Advanced 2D Virtual Try-On
 - Dynamic Scaling
 - Pose Estimation
 - Garment Alignment
 - Transparent PNG Assets
 - Cloth Segmentation
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10. Conclusion

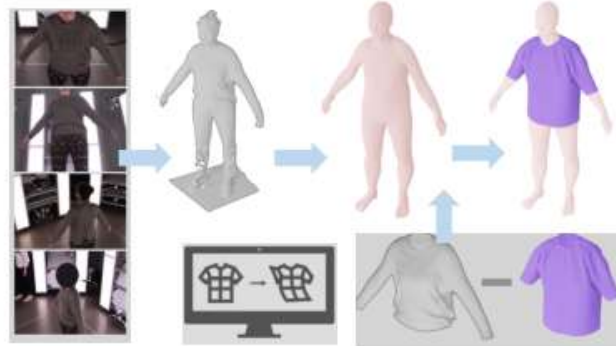
This research explored advanced 2D Virtual Try-On techniques that improve garment fitting and visual realism. Dynamic scaling, pose-based alignment, and transparent garment assets collectively create a better digital fitting experience.

(a) 2D virtual try-on



StableVITON

(b) 3D virtual try-on



DiffAvatar

(c) Image-based 3D virtual try-on



IPVTON